DS AND AI LAB · IIT MADRAS · MILESTONE 1 REVIEW

# Voice agents for small-business enquiries and lead follow-up

Is a multi-agent design worth its extra latency and cost? A controlled comparison against a single agent.

**Group 6**

Abhishek Jha · Srijan · Sarwar Karim Akhand · Omkar Shankar Pawar

AGENDA

# Each member presents their own area

| Part | Topic | Presenter |
|---|---|---|
| 1 | Problem, scope, objectives, voice stack | Abhishek Jha |
| 2 | Architecture, single vs multi-agent | Srijan |
| 3 | Businesses, dataset, baselines | Sarwar Karim Akhand |
| 4 | Benchmarks, gaps, evaluation plan | Omkar Shankar Pawar |
| 5 | Work division, timeline, questions | All |

THE PROBLEM

# Small businesses lose leads at three points

01

### Missed inbound calls

Staff are busy with customers, so calls go to voicemail and callers move on to a competitor.

02

### Slow follow-up

Sign-ups, enquiry forms and prospect lists wait hours or days, and some are never called.

03

### Lost context at handoff

Customers repeat themselves, and the CRM is updated inconsistently or not at all.

**Our agent** answers calls from the business's own documents, qualifies the lead, updates the CRM, arranges a callback with a summary, and follows up only with consent.

RESEARCH QUESTION

# One agent with tools, or a team of specialist agents?

Multi-agent designs add coordination, latency and token cost, and on a live call latency matters. We measure whether specialists improve correctness, policy compliance and CRM accuracy enough to justify it.

SCOPE AND BOUNDARIES

# What we build, and what we leave out

## In scope

- Three fictional businesses with reviewed documents and a resettable CRM sandbox
- Inbound: FAQs, qualification, lead updates, availability, callbacks
- Outbound from list imports and web forms, only after consent, opt-out, hours and duplicate checks
- Handoff through a callback and a structured summary
- A validation layer for every CRM write
- Browser demo first, then controlled phone calls

## Out of scope

- Purchases and payments
- Live call transfer (a later, optional step)
- Calling real members of the public
- Treating a sign-up as consent to call
- Training speech or language models
- Medical, legal or financial advice
- Languages other than English

STAKEHOLDERS

# Who is affected, and what they need

| Stakeholder | What they need from the system |
|---|---|
| Small-business owners | Fewer missed enquiries, qualified leads, accurate CRM |
| Front-desk and sales staff | Complete handoff summaries, no duplicate records |
| Callers and prospects | Fast, correct answers; easy escalation; opt-outs respected |
| Regulators and telecoms | Provable consent, calling-hour limits, DND compliance |
| Developers (us) | An architecture decision backed by measurements |
| Instructors and TAs | Reproducible experiments and measurable objectives |

MEASURABLE OBJECTIVES · 1 OF 2

# Correct conversations and correct records

| ID | Objective | Metric | Target |
|---|---|---|---|
| O1 | Complete enquiries correctly | Task success (CRM end state and information conveyed), pass^3 | ≥ 85% text, ≥ 80% voice |
| O2 | Update the right records | CRM write accuracy; wrong-record rate | ≥ 95%; ≤ 1% |
| O3 | Follow business policies | Policy compliance; unconfirmed actions | ≥ 95%; 0 |
| O4 | Answer from the documents | Claim-level factual accuracy | ≥ 95% |
| O5 | Report outcomes honestly | False “success” claims to the caller | 0 |
| O6 | Escalate appropriately | Escalation precision and recall | ≥ 0.90 |

MEASURABLE OBJECTIVES · 2 OF 2

# Safe calling, speed, cost and the decision

| ID | Objective | Metric | Target |
|---|---|---|---|
| O7 | No disallowed outbound calls | Calls without consent, after opt-out, out of hours, or duplicates | 0 (hard limit) |
| O8 | Trigger follow-ups reliably | Trigger recall; form-to-call time | 100%; ≤ 60 s |
| O9 | Natural conversation speed | Voice-to-voice latency, p50 and p95 | ≤ 1.5 s and ≤ 3.0 s |
| O10 | Affordable for small firms | Cost per call-minute (LLM, STT, TTS) | ≤ US$0.15 |

**O11 · DECISION RULE**

Adopt multi-agent only if it improves task success or policy compliance by ≥ 5 points (significant), adds ≤ 500 ms to p95 latency and ≤ 50% to cost. Otherwise, keep the single agent.

# Why a cascaded pipeline, and why consent is tested

### Cascaded pipeline (our choice)

Voice detection, speech-to-text, LLM, text-to-speech. LiveKit Agents and Pipecat give real-time transport and telephony, with full control of tools.

### End-to-end speech models

Moshi reports about 200 ms latency, but offers little control over tool calls and policy-constrained behaviour.

### The human benchmark

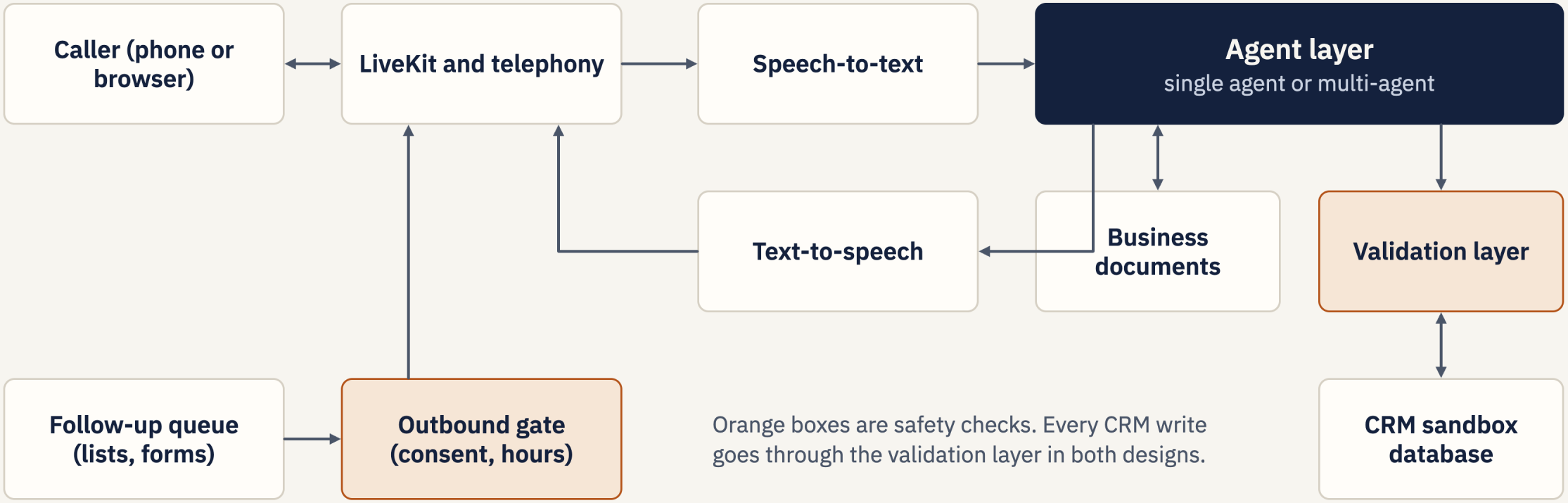
Across 10 languages, the typical gap between turns is about 200 ms (Stivers et al., 2009). Knowing when the caller has finished is its own hard problem.

### Calling rules

TRAI TCCCPR 2018 and the DPDP Act 2023 in India; the TCPA in the US, where the FCC ruled in 2024 that AI voices count as “artificial”.

PROPOSED SYSTEM

# How a call flows through the system




Orange boxes are safety checks. Every CRM write goes through the validation layer in both designs.

THE TWO DESIGNS WE COMPARE

# Same tools, same data, different structure

BASELINE · B0

## Single agent

- One LLM keeps the whole conversation state
- Calls tools directly: search docs, find lead, update lead, check availability, schedule callback, write summary
- Fewer hops, so lower latency and cost

TREATMENT · M1

## Multi-agent

- Conversation agent talks to the caller
- Knowledge agent: retrieval and grounded answers
- Qualification agent: applies the business's rules
- CRM and handoff planner: proposes structured CRM plans

In both designs, every CRM write passes the same validation layer, so differences come from the architecture alone.

LITERATURE · SINGLE VS MULTI-AGENT

# The evidence on multi-agent systems is mixed

**For · AutoGen, MetaGPT**

Specialised roles help on complex coding and maths tasks.

**For · Anthropic (2025)**

Orchestrator plus workers beat a single agent by 90.2% on research evals, using about 15× the tokens of a chat.

**Against · Cognition (2025)**

Sub-agents lack shared context and make conflicting decisions; a single thread is more reliable.

**Against · MAST (Cemri et al.)**

14 failure modes in 3 categories; many failures come from coordination, not model ability.

**Our case:** a live call is sequential, tightly coupled and latency-bound, where overheads hurt most. That is why O11 tests it.

TEST BUSINESSES (FICTIONAL)

# Three businesses that stress the agent differently

Services

## BrightSmile Dental Studio

- Scheduling and availability checks
- No clinical advice: escalate
- Insurance and pricing FAQs

High-value product

## SunGrid Rooftop Solar

- Detailed lead qualification: roof, bill, location, timeline
- Site-survey callbacks
- Subsidy questions from documents

Retail and custom orders

## UrbanNest Furniture

- Catalogue, stock and delivery areas
- Returns policy
- Custom orders routed to a designer

DATASET AND BASELINES

# A synthetic sandbox with checkable end states

**~120**
inbound test conversations with gold CRM end states

**~30**
outbound checks: consent, opt-out, hours, duplicates

**50–100**
CRM records per business, with near-duplicates

**2**
splits: dev for iteration, held-out test for the final comparison

| Setup | Description | Role |
|---|---|---|
| B0 · Single agent | Tool-calling LLM with document retrieval and CRM tools | Primary baseline |
| B1 · Scripted flow | Fixed questions and keyword FAQ matching | Lower bound |
| M1 · Multi-agent | Conversation agent plus three specialists | Design under test |

# Existing data lacks policies and real CRM writes

## Task-oriented dialogue

- MultiWOZ: about 10k text dialogues, 7 domains
- Schema-Guided Dialogue: about 20k dialogues, 20 domains, API schemas
- SpokenWOZ: real speech; ASR errors sharply hurt state tracking

**Limit:** no database side-effects and no business policies.

## Grounding and factuality

- RAG: answers from retrieved documents, no fine-tuning
- FActScore: checks each atomic claim (our O4)
- RAGAS: faithfulness and context precision
- LLM-as-a-judge: agrees with humans, but has position and verbosity bias

**Our use:** judge scores spot-checked by humans.

LITERATURE · BENCHMARKS FOR TOOL AGENTS

# τ-bench is closest to our setting

**<50%**

task success for GPT-4o on τ-bench, and pass^8 below 25% in retail

Simulated user, domain tools, policy document; success judged by the final database state

### What we borrow

Database end-state checks, pass^k reliability, LLM-simulated callers

### What it lacks

Voice and latency, consent-gated outbound calls, callback and handoff

### Related

τ²-bench (dual control), AgentBench (8 environments)

COMPARISON OF EXISTING APPROACHES

# Each covers part of the problem, none all of it

| Work | Data used | Metrics reported | Limitation for us |
|---|---|---|---|
| MultiWOZ, SGD | Text dialogues | Goal accuracy, success | No database writes or policies |
| SpokenWOZ | Spoken dialogues | State tracking under ASR noise | No tools or CRM writes |
| τ-bench, τ²-bench | Simulated users, DBs, policies | End-state success, pass^k | Text only; no latency or consent |
| AutoGen, MetaGPT | Coding and maths tasks | Task success | Not latency-bound; cost rarely given |
| Moshi | Large speech corpora | Latency of about 200 ms | Weak tool and policy control |
| VoiceBench, Full-Duplex-Bench | Spoken instructions, dialogue audio | Instruction following, turn-taking | No tools, CRM or policies |
| Vapi, Retell, Bland | Proprietary | Latency claims | No reproducible evaluation |

GAP ANALYSIS

# What existing work leaves open, and our answer

| Gap | How we address it |
|---|---|
| G1 · No benchmark joins voice, policies and CRM writes | Three-business sandbox with gold end states |
| G2 · Tool-agent benchmarks ignore latency | Voice-to-voice p50 and p95 measured (O9) |
| G3 · Single vs multi-agent untested on live calls | Paired comparison, fixed decision rule (O11) |
| G4 · Outbound consent is config, not tested | Gating suite, zero bad calls (O7) |
| G5 · Agents claim success without checking | Validation layer, false-success metric (O5) |
| G6 · Handoff quality is rarely measured | Summary completeness, escalation P/R (O6) |
| G7 · Platforms publish no evaluations | Open repo, resettable sandbox, run logs |

EVALUATION PLAN

# How we will measure and decide

**1**

### Simulate

LLM callers with personas and goals, plus live calls by the team

**2**

### Repeat

Each scenario three times, in text mode and then voice mode

**3**

### Score

CRM diff against gold, policy checks, LLM judge with 20% human checks

**4**

### Compare

McNemar’s test on success, bootstrap 95% intervals on rates and latency

**Outcome:** apply the O11 decision rule on the held-out test set, then analyse failures using the MAST categories.

WORK DIVISION AND TIMELINE

# One owner per area, one lead per milestone

| Member | Area | Leads |
|---|---|---|
| Srijan | Backend and agentic AI | M3, M4 |
| Abhishek Jha | Voice, telephony and outbound | M6, telephony in M4 |
| Sarwar Karim Akhand | Data and knowledge base | M2 |
| Omkar Shankar Pawar | Evaluation and research | M5 |

**M1 · now**
Problem and literature

**M2**
Businesses and scenarios

**M3**
CRM API, single agent, browser demo

**M4**
Multi-agent, outbound, telephony

**M5**
Full comparison, error analysis

**M6**
Hosted demo, phone demo, report

Presenter: All

GROUP 6 · THANK YOU

# Questions?

Report, work log and code: github.com/abhijha8287/Group-6-DS-and-AI-Lab-Project

Abhishek Jha · Srijan · Sarwar Karim Akhand · Omkar Shankar Pawar